

Configuration for Serial Number Recording

Requirements

The following prerequisites/restrictions are to be observed:

- In serial number recording, only a limited compensation of quantities is possible:
 - for serial number recording with scrap, no compensation with scrap must be configured for the yield quantity account.
 - for serial number recording with yield, no compensation with yield must be configured for the scrap quantity account.
- The entry of non-integer quantities is not permissible in connection with serial number recording.
- The serial number recording is only supported for yield or for scrap.
- Posting quantities is prevented by dialog control in the standard dialogs "Interrupt operation", "Log operation off" and "Partial confirmation" and/or "Output batch change", according to the machine mode. Quantities on OPs requiring serial numbers should therefore only be posted via the new dialog "A_SNR" serial number recording.

General

The type of serial number generation in HYDRA is primarily controlled by the "Serial no. obligation" identifier on the operation.

The identifier is set:

- directly on the operation, if the order/operation information from ERP are transferred or retrieved from a work plan.
- at the order type which then sets the option upon creation of an operation.

The following three options regarding serial number obligation are generally supported.

<i>Serial number obligation option on operation</i>	<i>Description</i>
E	Manual entry of serial number (on the basis of entered serial numbers on order) The serial number can be recorded by manual entry or bar code. A plausibility check regarding the order is performed.

	If the order also requires batch management, an output batch is generated (in the background) for the serial number and the serial number is assigned to the output batch.
G	Automatic generation of serial number The serial number is automatically generated through batch number assignment. A serial number corresponds to an output batch. - Can only be used with OPs requiring batch management - Is always posted with the Output batch change command
S	Generation of serial numbers using number ranges The serial number is requested and generated through the SNR number range at the HYDRA server - Can only be used with OPs requiring batch management - Serial number is assigned to the current output batch - Is always posted with the Output batch change command

Configuration with Orders not Requiring Batch Management

The configurations required for using and activating the setup function for the manual entry of serial numbers for orders/operations not requiring batch management on the AIP are described below.

Configuration on the Machine

The machine is **not** configured accordingly in batch mode.

The following identifiers are set in the workplace configuration → MPL:

- Batch management = N
- Automatic generation of batch number = N

Configuration on Order Type

In the order type of operations with batch number obligation, the dialog control = SNR must be entered.

The operations are not identified as requiring batch management.

For OPs not requiring batch management, the following settings are made on the OP with regard to "Serial no. obligation":

SERIENNR_PFLICHTIG	OPT:SNR	CHAR(1)	Serial number obligation Identifier which controls whether the order has serial number obligation and how it is entered/processed: "E" = Manual entry of SNR From which OP the serial number has to be entered is controlled by the "Seriennummern_pflichtig" (Serial number obligation) identifier at the processing code.
SERIENNR_VERGABE	OPT:SNRVERG	CHAR(1)	Determines how serial numbers are assigned (this field is only relevant if SERIENNR_PFLICHTIG <> "Y"/"N"): P = Serial numbers are transferred by the PPS system

Configuration at Processing Code

Field	Acronym	Type	Description
SERIENNR_PFLICHT	OPT:SNR=<>	CHAR (1)	Serial number obligation If this identifier is set, this means that from this operation on, postings must be by serial number. Y Yes, irrespective of whether serial numbers were transferred or not N No, even if serial numbers were transferred. Y/ N can thus be used to control from which operation an order requires a serial number.

Dynamic Dialog Configuration

The following configurations are prerequisite:

- Entry at the terminal is made via the dynamic dialog A_SNR which must exist in the system.
- The dialogs for Interrupt OP/Terminate OP (A_UN/A_AB, A_UN_MPL, A_AB_MPL) must be adapted in such a manner that the quantity fields are hidden by the dialog control SNR.

Layout Configuration at Terminal

Display of entered serial numbers in A_SNR dialog:

CTWINLAY.INI
<pre>[A_SNR.LST] GRID_FONT=Arial GRID_FONTSIZE=9</pre>

```
GRID_COLOR=clBlack
GRID_BACKGROUND=clWhite
GRID_ORDER=DAT=- | ZEI=- | SNR

EXAMINE_SCANEXPR1=BEL=G
EXAMINE_SCANCOLOR1=clGreen
EXAMINE_SCANEXPR2=BEL=A
EXAMINE_SCANCOLOR2=clRed
EXAMINE_SCANEXPR3=BEL=P
EXAMINE_SCANCOLOR3=clBlue
EXAMINE_SCANEXPR4=BEL=N
EXAMINE_SCANCOLOR4=clNavy

SNR=C20,90,L,Seriennummer
BEL=C1,30,Z,Kl.
GR=N8,40,R,Grund
GRTXT=C70,150,L,Grundtext
DAT=dd.mm.yyyy,70,Z,Datum
ZEI=hh:mm:ss,70,Z,Zeit
```

Display of existing serial numbers in the selection:

CTWINLAY.INI

```
[ Serial numbers ]
GRID_FONT=Arial
GRID_FONTSIZE=9
GRID_COLOR=clBlack
GRID_BACKGROUND=clWhite
GRID_ORDER=SNR

DY1=C1,50,Z
SNR=C20,250,L,Seriennummer
DY2=C1,10,Z
BEL=C1,50,Z,Klasse
```

Configuration with Orders Requiring Batch Management

The configurations required for using and activating the setup function for the manual entry of serial numbers or for the automatic generation/entry of serial numbers for orders/operations requiring batch management on the AIP are described below.

Configuration on the Machine

The machine is **not** configured in batch mode.

The following identifiers are set in the workplace configuration → MPL:

- Batch management = N
- Automatic generation of batch number = Y

Configuration on Order Type

In the order type of operations with batch number obligation, the dialog control = SNR must be entered.

As required, the serial number recording is configured accordingly in the order type:

- The SNR is to be entered manually and checked against the serial numbers entered in the order header. In addition, a batch for the serial number is to be created → Serial no. obligation identifier = "E"
- The SNR is to be generated automatically (HYDRA batch number) and entered and administered as a batch in HYDRA. → Serial no. obligation identifier = "G"
- The SNR is to be generated automatically and a batch assigned to the serial number (with a HYDRA batch number) is to be created and administered in HYDRA. The serial number is created in a specific number range. This number range can be configured freely through customizing. → Serial no. obligation identifier = "S"

For OPs requiring batch management, the following settings are made on the OP with regard to "Serial no. obligation":

SERIENNR_PFLICHTIG	OPT:SNR	CHAR(1)	Serial number obligation Identifier which controls whether the order has serial number obligation and how it is entered/processed: "G" = Autom. generation of SNR "S" = Autom. generation of SNR via number range "E" = Manual entry of SNR From which OP the serial number has to be entered is controlled by the "Seriennummern_pflichtig" (Serial number obligation) identifier at the processing code.
SERIENNR_VERGABE	OPT:SNRVERG	CHAR(1)	Determines how serial numbers are assigned (this field is only relevant if SERIENNR_PFLICHTIG <> " " / "N"): P = Serial numbers are transferred by the PPS system

Configuration at Processing Code

Field	Acronym	Type	Description
SERIENNR_PFLICHT	OPT:SNR=<>	CHAR (1)	Serial number obligation If this identifier is set, this means that from this operation on, postings must be by serial number.

Field	Acronym	Type	Description
			Y Yes, irrespective of whether serial numbers were transferred or not N No, even if serial numbers were transferred. Y/ N can thus be used to control from which operation an order requires a serial number.

Configuration at Material Type

Whether or not serial numbers are entered for operations requiring batch management is also set at the material type. If the option is inactive, serial numbers will not be recorded as batches.

Dynamic Dialog Configuration

The following configurations are prerequisite:

- Entry at the terminal is made via the dynamic dialog A_SNR which must exist in the system.
- The dialogs for Interrupt OP/Terminate OP (A_UN/A_AB, A_UN_MPL, A_AB_MPL) must be adapted in such a manner that the quantity fields are hidden by the dialog control SNR.

Layout Configuration at Terminal

Display of entered serial numbers in dialog A_SNR:

CTWINLAY.INI

```
[ A_SNR.LST ]
GRID_FONT=Arial
GRID_FONTSIZE=9
GRID_COLOR=clBlack
GRID_BACKGROUND=clWhite
GRID_ORDER=DAT=- | ZEI=- | SNR

EXAMINE_SCANEXPR1=BEL=G
EXAMINE_SCANCOLOR1=clGreen
EXAMINE_SCANEXPR2=BEL=A
EXAMINE_SCANCOLOR2=clRed
EXAMINE_SCANEXPR3=BEL=P
EXAMINE_SCANCOLOR3=clBlue
EXAMINE_SCANEXPR4=BEL=N
EXAMINE_SCANCOLOR4=clNavy

SNR=C20,90,L,Seriennummer
BEL=C1,30,Z,K1.
GR=N8,40,R,Grund
```

```
GRTXT=C70,150,L,Grundtext  
DAT=dd.mm.yyyy,70,Z,Datum  
ZEI=hh:mm:ss,70,Z,Zeit
```

Display of existing serial numbers in the selection:

CTWINLAY.INI

```
[ Serial numbers ]  
GRID_FONT=Arial  
GRID_FONTSIZE=9  
GRID_COLOR=clBlack  
GRID_BACKGROUND=clWhite  
GRID_ORDER=SNR  
  
DY1=C1,50,Z  
SNR=C20,250,L,Seriennummer  
DY2=C1,10,Z  
BEL=C1,50,Z,Klasse
```